

VoltCore ERP

Enterprise Resource Planning System for Power Plant EPC Contractors

VoltCore Engineering Pvt. Ltd.
Version 1.0.0

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1. Executive Summary

VoltCore ERP is a comprehensive, full-featured Enterprise Resource Planning system purpose-built for Indian power plant EPC (Engineering, Procurement, and Construction) contractors. Developed with a modern technology stack comprising Next.js 16, TypeScript, TailwindCSS 4, and Prisma ORM, the system delivers a unified platform for managing every aspect of construction-project operations from human resources and finance to inventory control and site safety compliance.

The application encompasses 12 functional modules covering HRMS, Finance, Projects, Inventory, Operations, Sales, CRM, Support, Knowledgebase, and more. At the data layer, VoltCore ERP leverages 34 database tables across SQLite (development) and 63 MySQL tables (production), populated with over 1,300 seed data records for immediate demonstration and testing. The backend exposes 33 RESTful API endpoints with full CRUD operations (GET, POST, PUT, DELETE, PATCH), while the frontend comprises 25+ dynamically-loaded module components.

The user interface features a dark industrial theme with an amber (#f5a623) accent color, carefully designed for readability during extended use in field and office environments. Built on the shaden/ui component library, the interface is fully mobile-responsive with adaptive layouts for desktop, tablet, and mobile viewports. Zustand manages client-side navigation state, enabling seamless single-page routing across all modules without full-page reloads.

VoltCore ERP maintains a zero-lint codebase with comprehensive error handling, input validation, and auto-generated unique identifiers for all entity types (PRJ-XXX, EMP-XXX, PO-XXXX, INV-XXXX). The system includes auto-computed statutory deductions (PF, ESI, TDS), double-entry bookkeeping, budget variance tracking, shift roster management, and permit-to-work workflows, making it a production-ready solution for the demanding requirements of Indian EPC contracting.

2. System Architecture

2.1 Frontend Architecture

- Next.js 16 App Router with a single-page ERP layout hosting all 12 functional modules.
- Zustand store for global navigation state including active module routing and sidebar collapsed/expanded mode.
- Dynamic imports for code splitting across 25+ module components, reducing initial bundle size.
- Dark industrial theme with custom CSS utility classes and amber (#f5a623) accent throughout.
- Recharts library for interactive data visualization dashboards across finance, HRMS, and project modules.

2.2 Backend Architecture

- 33 RESTful API routes organized under /api/*, each handling GET, POST, PUT, DELETE, and PATCH methods.
- Prisma ORM with SQLite for rapid development and MySQL 8.0+ for production deployments.
- Full CRUD operations on all entities with auto-generated unique IDs (PRJ-XXX, EMP-XXX, PO-XXXX, INV-XXXX).
- Comprehensive input validation and try/catch error handling returning structured JSON responses.
- Structured seed data (1,300+ records) enabling immediate demonstration upon deployment.

2.3 Database Design

Metric	Count
Total Tables (SQLite/Prisma)	34
API Routes	33
Frontend Modules	25+
Seed Data Records	1,300+
MySQL Tables (Production v2)	63

3. Module Reference

3.1 Organization

The Organization module establishes the foundational corporate hierarchy, defining departments, designations, role structures, and associated salary bands that cascade through all other modules.

Sub-Module	Features	Database Tables
Departments & Roles	Department hierarchy, role definitions, salary bands, organizational chart	Department, Designation

3.2 HRMS (Human Resources)

The HRMS module provides end-to-end workforce management from onboarding through payroll processing. It tracks 30 employees with 957 attendance records, auto-computes statutory deductions including PF, ESI, and TDS, and manages shift planning, leave approval workflows, training certifications, and recruitment pipelines.

Sub-Module	Features
Employee Analytics	Headcount trends, department distribution, demographic charts
Employees	Full employee profiles, designation, department, salary, contact info
Attendance	957 attendance records, daily clock-in/out, late tracking
Leave Management	Leave types, balance tracking, approval workflow
Shift Roster	Shift planning, rotation schedules, crew assignments
Timesheet	Weekly/biweekly timesheets, overtime calculation, project allocation
Payroll	Auto-computed PF, ESI, TDS; salary slips, payslip generation
Training & Certifications	Certification tracking, expiry alerts, training schedules
Recruitment	Job postings, applicant tracking, interview stages

3.3 Procurement

The Procurement module manages the complete purchase lifecycle from requisition through Goods Received Note (GRN) tracking, along with expense claim approval workflows and vendor management.

Sub-Module	Features
Purchase Orders	PO lifecycle (created/approved/ordered/received), GRN tracking, vendor management
Expenses	Expense claim submission, approval workflows, category tracking

3.4 Finance

The Finance module implements double-entry bookkeeping with 22 ledger accounts, accounts payable and receivable, journal entries, bank and cash management, tax compliance, budget planning with variance tracking, and comprehensive financial reporting.

Sub-Module	Features
Dashboard	Financial KPIs, cash flow summary, key metrics at a glance
Ledger	22 ledger accounts, chart of accounts, double-entry posting
Accounts Payable	Vendor bills, payment scheduling, aging reports
Accounts Receivable	Customer invoices, payment tracking, follow-up alerts
Journal Entries	Manual journal entries, debit/credit posting, batch entries

Sub-Module	Features
Bank & Cash	Bank reconciliation, cash flow tracking, multiple accounts
Taxation	GST computation, tax filing support, compliance tracking
Budget	Budget allocation, variance analysis, period-over-period comparison
Financial Reports	P&L, balance sheet, cash flow statement, trial balance

3.5 Projects

The Projects module tracks 8 active projects across India with progress monitoring, multi-site management, and site mapping capabilities for power plant construction oversight.

Sub-Module	Features
All Projects	8 projects across India, progress tracking, status monitoring, timelines
Site Map	Multi-site management, geographic project locations, site details

3.6 Inventory

The Inventory module manages 20 stocked items with 35 recorded movements, low stock alerts, and warehouse-level tracking for materials and equipment used in construction activities.

Sub-Module	Features
Items	20 inventory items, categories, unit pricing, stock levels
Stock Movements	35 movement records, inbound/outbound tracking, transaction history

3.7 Assets & Operations

The Assets & Operations module handles equipment lifecycle management, Permit-to-Work (PTW) workflows, safety and HSE incident tracking, and subcontractor compliance documentation.

Sub-Module	Features
Equipment	Equipment registry, lifecycle tracking, maintenance schedules
Work Permits	Permit-to-Work (PTW) management, approval workflow, type classification
Safety & HSE	Safety incident tracking, near-miss reporting, HSE compliance metrics
Subcontractors	Subcontractor registration, compliance documents, performance tracking

3.8 Sales

The Sales module provides customer relationship management with order pipeline tracking and GST computation for invoicing and revenue management.

Sub-Module	Features
Customers	Customer profiles, contact details, GST identification
Sales Orders	Order pipeline, status tracking, GST computation, order history

3.9 CRM

The CRM module manages the sales pipeline with lead tracking through deal stages, source attribution, and contact management for business development activities.

Sub-Module	Features
Contacts	Lead pipeline, deal stages, source tracking, contact management

3.10 System

The System module provides centralized report generation and application-wide configuration including company settings, user preferences, and multi-format report exports.

Sub-Module	Features
Reports	Multi-report generation across modules, export capabilities
Settings	Company configuration, user preferences, system parameters

3.11 Support

The Support module delivers priority-based ticket management with resolution tracking, ensuring timely handling of internal and external support requests.

Sub-Module	Features
Tickets	Priority-based ticket management, status tracking, resolution workflow

3.12 Knowledgebase

The Knowledgebase module provides a centralized repository of categorized articles with tags, helpful ratings, and view counts for organizational knowledge sharing.

Sub-Module	Features
Articles	Categorized articles, tagging, helpful ratings, view counts

4. Technology Stack

VoltCore ERP is built on a modern, high-performance technology stack chosen for developer productivity, type safety, and production reliability.

Layer	Technology	Version
Frontend Framework	Next.js	16.1.3
Language	TypeScript	5.x
Styling	Tailwind CSS	4.x
UI Components	shadcn/ui	Latest
State Management	Zustand	5.x
ORM	Prisma	6.x
Charts	Recharts	2.x
Icons	Lucide React	Latest
Animations	Framer Motion	12.x

5. Setup & Deployment

5.1 Prerequisites

- Node.js 18+ or Bun runtime installed on the development machine.
- MySQL 8.0+ for production database (SQLite 3 used for development).
- Git for version control and repository management.

5.2 Development Setup

Follow these steps to set up the VoltCore ERP development environment:

1. Clone the repository: `git clone <repository-url>`
2. Copy environment configuration: `cp .env.example .env`
3. Configure DATABASE_URL in the .env file

4. Install dependencies: `bun install`
5. Push database schema: `bun run db:push`
6. Seed the database: `bun run db:seed`
7. Start the development server: `bun run dev`
8. Access the application: `http://localhost:3000`

5.3 Database Migration to MySQL

For production deployment, VoltCore ERP supports migration from SQLite to MySQL 8.0+ with 63 tables including enhanced indexing and constraints.

1. Import MySQL schema: `mysql -u root -p voltcore_erp < database/voltcore_erp_mysql.sql`
2. Update `.env`: `DATABASE_URL=mysql://user:pass@host:3306/voltcore_erp`
3. Generate Prisma client: `bun run db:generate`
4. Push schema changes: `bun run db:push`

6. File Structure

The following table maps the key directories and files in the VoltCore ERP project repository.

Path	Description
<code>src/app/page.tsx</code>	Main ERP page with module routing and layout
<code>src/app/api/*/route.ts</code>	API route handlers (33 endpoints)
<code>src/components/erp/*.tsx</code>	Module UI components (25+ components)
<code>src/store/erp-store.ts</code>	Zustand navigation store for sidebar and module state
<code>prisma/schema.prisma</code>	Database schema definition with 34 models
<code>prisma/seed.ts</code>	Database seeder script (1,300+ records)
<code>database/voltcore_erp_mysql.sql</code>	MySQL DDL + seed data (63 tables)
<code>download/</code>	Exportable project files and documentation